

AI engineering bootcamp

This course covers the end-to-end lifecycle of an AI product. Students begin with high-level Python and Statistics to understand the "math of meaning." They then progress through building predictive models, computer vision systems, and advanced Retrieval-Augmented Generation (RAG) pipelines. The final phase focuses on Agentic AI, creating autonomous systems that can use tools and work in teams, culminating in MLOps strategies for deploying these systems to the cloud using Docker and FastAPI.

Who Can Join the Course

- **Software Developers:** Looking to pivot into AI without needing a PhD.
- **Data Analysts:** Who want to move from descriptive analytics to building autonomous AI tools.
- **Computer Science Students/Grads:** Seeking hands-on, industry-standard project experience.
- **Tech Professionals:** Interested in "Sovereign AI" (private, self-hosted solutions for enterprise).
- **AI enthusiast:** Interested in AI development and implementing AI with real world use cases

Total Live & Support Classes

- **Live Classes:** 40 Sessions (2 per week, 1.5 hours each).
- **Support Classes:** 40 Sessions (2 per week, focused on debugging, local environment setup, and assignments).

Total Number of Projects

- 16 Hands-on Projects

Course Prerequisites

- **Basic Programming Logic:** Familiarity with concepts like loops, variables, and functions in any language.
- **Mathematics Intuition:** Basic high-school-level understanding of algebra and probability (though we teach the AI-specific parts).
- **Hardware Requirements:** A computer with at least 8GB RAM (16GB recommended) to run small local models.
- **Tooling:** Willingness to work with the Command Line (CLI) and VS Code.

Month 1: Foundations of Data & Logic

Week 1: The AI Workbench & Python Essentials

- **Class 1** (Module 1)
 - **Focus Area:** Data Foundations (Setup & Syntax).
 - **Topics:** AI Engineer vs. ML Engineer roles; Environment setup (Python `venv`, Conda, VS Code); Local LLM setup with **Ollama**; Python data types and control flow.
 - **Achievements:** Run a LLM model locally and automate a text-sorting task using Python.
- **Class 2** (Module 1)
 - **Focus Area:** Advanced Python for AI.
 - **Topics:** List comprehensions; Lambda functions; Handling JSON/YAML for model configs; Error handling for API failures.
 - **Achievements:** Build a local Python script that parses raw data files into a clean format.

Week 2: Numerical Foundations

- **Class 3** (Module 1)
 - **Focus Area:** Numerical Processing with NumPy.
 - **Topics:** N-dimensional arrays; Vectorization vs. Loops; Dot products and Matrix multiplication (The language of AI).
 - **Achievements:** Build a "Similarity Engine" using NumPy to compare the distance between data points.
- **Class 4** (Module 1)
 - **Focus Area:** Data Orchestration with Pandas.
 - **Topics:** DataFrames and Series; Data cleaning (Handling NaN, Outliers); GroupBy operations; Merging private datasets.
 - **Achievements:** Perform a complete audit on a "dirty" row dataset.

Week 3: Visualization & Descriptive Statistics

- **Class 5** (Module 1 & 2)
 - **Focus Area:** Visual EDA & Descriptive Stats.
 - **Topics:** Matplotlib & Seaborn; Mean, Median, Mode; Standard Deviation and Variance; Distribution analysis (Histograms/Boxplots).
 - **Achievements:** Create a "Data Storyboard" showing the health and trends of a business dataset.
- **Class 6** (Module 2)

- **Focus Area:** Statistics for AI (Probability).
- **Topics:** Normal Distribution; Z-scores; Probability Theory (Bayesian vs. Frequentist); Why it matters for model confidence.
- **Achievements:** Calculate the "Confidence Score" of a simulated prediction system.

Week 4: Statistical Testing & Inferential Stats

- **Class 7** (Module 2)
 - **Focus Area:** Statistics for AI (Hypothesis Testing).
 - **Topics:** Central Limit Theorem; P-values and Significance; T-tests and ANOVA; Correlation vs. Causation.
 - **Achievements:** Conduct an A/B test analysis to determine if a prompt change actually improved model accuracy.
 - **Class 8** (Module 2)
 - **Focus Area:** Real-world Project
 - **Topics:** End-to-end statistical analysis of a company dataset; Finding actionable insights.
 - **Achievements:** Deliver a "Statistical Insight Report" for a mock dataset.
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Month 2: Machine Learning & Deep Learning

Week 5: Supervised Learning - Regression

- **Class 9** (Module 3)
 - **Focus Area:** Machine Learning (Predicting Values).
 - **Topics:** Linear Regression; Gradient Descent (simplified); Mean Squared Error (MSE); Feature Scaling with Scikit-Learn.
 - **Achievements:** **Project: House Price Predictor** (A local tool to estimate real estate value).
- **Class 10** (Module 3)
 - **Focus Area:** Supervised Learning (Basic Classification Model).
 - **Topics:** Logistic Regression; Decision Trees; Random Forests; Feature importance.
 - **Achievements:** Build a "**Loan Approval Predictor**" that explains why it approved or denied a user.

Week 6: Machine Learning - Classification II

- **Class 11** (Module 3)
 - **Focus Area:** Machine Learning (Advanced Classification Model).

- **Topics:** Naive Bayes; Support Vector Machines (SVM); K-Nearest Neighbors (KNN).
 - **Achievements:** Build a local "**Spam & Intent Classifier**" that runs 100% offline.
- **Class 12** (Module 3)
 - **Focus Area:** Unsupervised Learning.
 - **Topics:** K-Means Clustering; Principal Component Analysis (PCA); Market Basket Analysis.
 - **Achievements:** Create a "**Customer Segmentation Engine**" that groups customers based on purchasing behavior.

Week 7: Deep Learning Foundations

- **Class 13** (Module 4)
 - **Focus Area:** Neural Networks.
 - **Topics:** The Neuron; Weights and Biases; Activation Functions (ReLU, Sigmoid); Backpropagation (The "Intuition").
 - **Achievements:** Build a simple 2-layer neural network from scratch in Python to solve a logic gate problem.
- **Class 14** (Module 4)
 - **Focus Area:** Tensors & Frameworks.
 - **Topics:** Introduction to TensorFlow/Keras; Tensor operations; Building sequential models; Loss functions.
 - **Achievements:** Train a deep learning model to predict a "**Success Score**" for industrial equipment.

Week 8: Advanced Deep Learning

- **Class 15** (Module 4)
 - **Focus Area:** Deep Learning (CNNs).
 - **Topics:** Convolutional layers; Max Pooling; Padding and Stride; Image normalization.
 - **Achievements: Project: MNIST Digit Classifier** (A system that recognizes handwritten numbers).
- **Class 16** (Module 4)
 - **Focus Area:** Deep Learning (Optimization).
 - **Topics:** Overfitting vs. Underfitting; Dropout layers; Batch Normalization; Early Stopping.
 - **Achievements:** Optimize your MNIST classifier to reach >98% accuracy.

Month 3: NLP, Vision & Generative AI

Week 9: Computer Vision & OpenCV

- **Class 17** (Module 5)
 - **Focus Area:** Image Processing.
 - **Topics:** OpenCV basics; Image Kernels (Blur, Sharpen); Grayscale and Thresholding; Contour detection.
 - **Achievements:** Create a "**Document Scanner**" script that fixes the perspective of a photo.
- **Class 18** (Module 5)
 - **Focus Area:** Vision Models in Practice.
 - **Topics:** Pre-trained Vision models (YOLO, ResNet); Real-time object detection using a webcam.
 - **Achievements:** Build a "**Privacy Shield**" that automatically blurs human faces in a live video stream.

Week 10: Natural Language Processing (NLP)

- **Class 19** (Module 5)
 - **Focus Area:** NLP Basics & Transformers.
 - **Topics:** Tokenization (BPE); Word Embeddings (Word2Vec/GloVe); The Transformer "Attention" mechanism.
 - **Achievements:** Build a "**Keyword Matcher**" that finds relevant CVs for a job description based on meaning.
- **Class 20** (Module 5)
 - **Focus Area:** Text Classification.
 - **Topics:** Fine-tuning a BERT/DistilBERT model for sentiment analysis; HuggingFace Transformers library.
 - **Achievements:** Build a local "**Customer Feedback Analyzer**" that labels reviews as Positive, Negative, or Neutral.

Week 11: Generative AI & Prompt Engineering

- **Class 21** (Module 6)
 - **Focus Area:** LLM Integration & Prompting.
 - **Topics:** Zero-Shot vs. Few-Shot; Chain-of-Thought (CoT); Prompt Injection defenses; Temperature vs. Top-P.
 - **Achievements:** Build a "**Brand Content Generator**" that strictly follows a company's unique writing style.
- **Class 22** (Module 6)
 - **Focus Area:** Structured Outputs & Frameworks.
 - **Topics:** Getting JSON from LLMs; Introduction to **LangChain**; Prompt Templates; Chains of reasoning.
 - **Achievements:** Build a "**Resume to JSON**" converter that extracts experience and skills into a database format.

Week 12: Retrieval-Augmented Generation (RAG)

- **Class 23** (Module 6)
 - **Focus Area:** Vector Databases.
 - **Topics:** Setting up **ChromaDB** locally; Semantic Search vs. Keyword Search; Text Chunking strategies.
 - **Achievements:** Index a 200-page employee handbook into a local vector database.
 - **Class 24** (Module 6)
 - **Focus Area:** RAG Pipelines.
 - **Topics:** Combining LLMs with Vector DBs; The "Open-Book Exam" analogy; Metadata filtering.
 - **Achievements:** Creating A bot that answers questions strictly from your uploaded PDFs.
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Month 4: Agentic AI & Production MLOps

Week 13: Agentic AI Fundamentals

- **Class 25** (Module 7)
 - **Focus Area:** Tool Calling & Autonomous Logic.
 - **Topics:** Giving AI "Hands"; Function Calling with local models (Mistral/Llama 3.1); Writing Python tools for agents.
 - **Achievements:** Build an AI that can calculate math or check your local file system to answer questions.
- **Class 26** (Module 7)
 - **Focus Area:** The ReAct Loop.
 - **Topics:** Reasoning and Acting loops; Multi-step task decomposition; Managing agent failures.
 - **Achievements:** Build a "**Travel Agent**" that can search a local database of flights and book one (simulated).

Week 14: Multi-Agent Systems

- **Class 27** (Module 7)
 - **Focus Area:** Orchestration with LangGraph.
 - **Topics:** State Management; Nodes and Edges; Human-in-the-loop (Approval steps).
 - **Achievements:** Build a "**Self-Correcting Writer**" that writes code, tests it, and fixes it if it fails.
- **Class 28** (Module 7)
 - **Focus Area:** Agentic Workflows (CrewAI).
 - **Topics:** Defining Roles and Hierarchies; Multi-agent collaboration; Sequential vs. Parallel tasks.

- **Achievements:** Creating a multi-agent team that researches a topic and writes a full report.

Week 15: MLOps & Serving

- **Class 29** (Module 8)
 - **Focus Area:** Model Serving with FastAPI.
 - **Topics:** Wrapping AI agents in REST APIs; Async serving; Throughput vs. Latency.
 - **Achievements:** Turn your **Question Answering Bot** project into a professional web service with a live API documentation page.
- **Class 30** (Module 8)
 - **Focus Area:** Monitoring & Observability.
 - **Topics:** Tracking token usage and costs; Monitoring for "Drift" and "Hallucinations" with **Langfuse**.
 - **Achievements:** Set up a dashboard that tracks every query's cost, time, and quality score.

Week 16: Advanced Agentic Industry Solutions part 1

- **Class 31 & 32** (Module 9)
 - **Focus Area:** Build a Financial AI & Automated Research Agent
 - **Topics:** Utilizing multi-agent orchestration with **CrewAI** for real-time market data extraction; Performing semantic sentiment analysis on financial news; Generating automated PDF investment memos with a focus on local data persistence.
 - **Achievements:** Build a system where a **"Researcher Agent"** finds stock news and a **"Analyst Agent"** calculates risk scores entirely offline.

Week 17: Advanced Agentic Industry Solutions part 2

- **Class 33 & 34** (Module 9)
 - **Focus Area:** Privacy-First Healthcare RAG
 - **Topics:** Implementing a local RAG system using vector DB; Local model.
 - **Achievements: Graduate** Build a healthcare assistant that can answer medical queries from a local database

Week 18: Project Deployment Strategies

- **Class 35 & 36** (Module 10)
 - **Focus Area:** Dockerization & Cloud.
 - **Topics:** Containerizing the AI Stack (UI + DB + LLM); GPU passthrough in Docker; Deployment to cloud server.
 - **Achievements:** Create a **"Single-Click Deployment"** for your entire AI application.

Week 19: Capstone Project

- **Class 37 & 38** (Module 10)
 - **Focus Area:** Capstone Project
 - **Topics:** Standard project management methodologies; Capstone selection guidelines; Fundamentals of project planning and execution.
 - **Achievements:** Based on the selected topic student will develop a end to end solution of a real problem by maintaining proper project management methodologies

Week 20: AI Engineer Job Readiness & Resume Building

- **Class 39** (Module 10)
 - **Focus Area:** Capstone Presentation.
 - **Topics:** Project Showcasing Day
- **Class 40**
 - **Focus Area:** Capstone Presentation.
 - **Topics:** GitHub Portfolio guidance; AI Engineer Resume Building; Industry Interview Strategy.
 - **Achievements:** Every student leaves with a finalized **AI Engineer Resume**, a polished **LinkedIn Profile**, and a **GitHub Portfolio** featuring 16 production-ready projects.